

Module 9

Technical Workouts

1. 1. Understand Exploratory Data Analysis (EDA)

- a. Purpose of EDA
- b. Understanding datasets
- c. Identifying patterns
- d. Detecting anomalies
- e. Data storytelling

2. 2. Understand Descriptive Analysis

- a. Summary statistics
- b. Distribution analysis
- c. Variability analysis
- d. Central tendency analysis
- e. Dataset profiling

3. 3. Understand Univariate Analysis

- a. Single-variable analysis
- b. Histograms
- c. Frequency distributions
- d. Outlier detection
- e. Distribution interpretation

4. 4. Understand Bivariate Analysis

- a. Feature relationships
- b. Correlation analysis
- c. Scatter plots
- d. Comparative analysis
- e. Trend identification

5. 5. Understand Multivariate Analysis

- a. Multiple feature relationships
- b. Feature interaction analysis
- c. Pattern discovery
- d. Dimensional thinking
- e. Business insights generation

6. 6. Understand Visualization Principles

- a. Choosing the right chart
- b. Data storytelling
- c. Visualization best practices
- d. Avoiding misleading visuals
- e. Effective communication

7. 7. Understand ML-Oriented EDA

- a. Target variable analysis
- b. Feature importance awareness
- c. Bias identification
- d. Data quality validation
- e. Preparing for model building